

ISRO, Space & Defence Tech

Launch vehicles, missions, missiles and dual-use space - defence literacy - rising theme.

Portion focus ~76% · Topic: sci-space-defence

PYQ themes: ISRO missions · Launch vehicles · Missiles · Space applications · Nuclear/defence awareness

Why this portion (PYQ lens)

Portion chance ~76% (rising). Parent sci-space-defence ~75%. Missions/missiles CA - static blend.

Years appeared (synthesis): 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025

Why UPSC asks this

Named missions and missile families create matching questions with rising frequency.

Syllabus framing

Portion chance ~76% (rising). Parent sci-space-defence ~75%. Missions/missiles CA - static blend. This portion sits inside sci-space-defence and centres on space missions and defence technology. Named missions and missile families create matching questions with rising frequency.

Recurring PYQ angles: ISRO missions, Launch vehicles, Missiles, Space applications, Nuclear/defence awareness. Approximate appearance years in the 2016-2025 educational synthesis: 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025. Use the chance percentage only to sequence study hours - it is not an official prediction.

Core map of space missions and defence technology

PSLV/GSLV/LVM3: Workhorse vs heavier lift narratives. SSLV: Small satellite launch niche.

Chandrayaan lineage: Lunar exploration cards - refresh latest. Mangalyaan: Mars orbiter cost-effectiveness lore.

Hold this core map as your elimination spine before memorising peripheral CA overlays.

Core concepts and definitions

Gaganyaan: Human spaceflight programme awareness. NavIC: Regional navigation - strategic autonomy.

Remote sensing: Resource/disaster applications. Satcom: Communication/edutainment/governance uses.

Institutions, processes and mechanisms

Space debris/safety: Emerging statement themes. Missile families: Agni/Prithvi/BrahMos - roles outline.

MRBM/IRBM/ICBM cues: Range class vocabulary. Missile defence sketch: Awareness level.

India angle and applied examples

Dual use: Space tech civilian + strategic. NSIL/IN-SPACe: Institutional reform headlines.

Private sector entry: Policy shift CA. ASAT test awareness: Strategic signalling literacy.

Comparisons, linkages and discriminators

Cryogenic tech: GSLV capability story. Earth observation constellations: Applications over alphabet soup.

Defence research bodies: DRDO headline link. Refresh cadence: Living list of last 24 months missions.

India angle

Maintain a 1-page living ISRO+missile sheet dated monthly before Prelims.

Question patterns and exam spine

Matching lists and statements on space missions and defence technology.

Revision spine: Definitions -> table -> India -> traps.

Eliminate anachronisms and category errors first, then test remaining options against table rows and must-remember triggers.

Trap watch-list (specific)

Avoid: Mixing PSLV/GSLV capabilities casually. Avoid: Outdated mission status. Avoid: Every satellite is navigation. Avoid: Range class mix-ups.

Vehicle cues

Vehicle | Cue

PSLV -- Workhorse SSO etc.

GSLV -- Heavier/geo-capable lineage

LVM3 -- Heavy lift

SSLV -- Small/quick

Map / list facts

- Sriharikota
- Downrange ocean safety narratives

Common traps

- x Mixing PSLV/GSLV capabilities casually
- x Outdated mission status
- x Every satellite is navigation
- x Range class mix-ups

Must remember

- PSLV vs GSLV/LVM3
- NavIC
- Chandrayaan/Gaganyaan refresh
- Missile family roles
- Applications > acronyms

Quick revision checklist

- [] Reproduce table
- [] 5 must-remembers
- [] 3 traps
- [] India anchor
- [] 10 MCQs

Educational prioritisation heuristic - not an official UPSC prediction. (c) UPSC IAS Study Hub original notes.